

Tewodros Amberbir Habtegebrial

PHD CANDIDATE · COMPUTER VISION

RPTU Kaiserslautern-Landau, Gottlieb-Daimler-Straße 47, 67663 Kaiserslautern

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Education

RPTU Kaiserslautern-Landau

Kaiserslautern, Germany

PHD, COMPUTER SCIENCE

Jan. 2018 - Dec. 2023

- Advisor: Prof. Didier Stricker
- PhD Thesis: "Learning Renderable 3D Representations from Minimal Scene Specifications".

RPTU Kaiserslautern-Landau

Kaiserslautern, Germany

MSC, COMPUTER SCIENCE

Nov. 2014 - Jan. 2017

- Advisors: Dr. Christian Bailer, Prof. Didier Stricker
- Thesis: "Novel-view Synthesis with Deep Convolutional Networks".
I developed a novel image-based rendering technique using Siamese patch-matching networks.
- Major subject: Intelligent Systems
- Minor subject: Psycho-linguistics
- Overall Grade: 1.3

Mekelle Institute of Technology

Mekelle, Ethiopia

BSC, ELECTRONICS AND COMMUNICATIONS ENGINEERING

Oct. 2007 - Aug. 2012

- Advisor: Dr. Workneh Gebeyehu
- Thesis: "Visual Object Tracking Using Particle Filtering".
I worked on video object tracking methods using Monte Carlo methods such as Particle Filters.
- Overall Grade: 3.79/4.00

Awards & Fellowships

Google PhD Fellowship

2021-2023

GOOGLE RESEARCH

in Machine Perception, Speech Technology and Computer Vision

CS Graduate Scholarship

2015-2021

RPTU KAISERSLAUTERN-LANDAU

Kaiserslautern, Germany

Graduate Scholarship to pursue Master in Computer Science

Merit Scholarship

2008-2012

MEKELLE INSTITUTE OF TECHNOLOGY

Mekelle, Ethiopia

Awarded to top 1% of highschool graduates

Internships and Student Researcher Positions

Google Research

Remote

STUDENT RESEARCHER: WITH DR. STEFAN POPOV

Aug. 2022 - Mar. 2023

- Topic: "Multi-view Consistent Diffusion Models"

Meta Reality Labs

Remote

INTERN: WITH DR. CHAKRAVARTY CHAITANYA, DR. FEDERICO PERAZZI

Jun. - Sep. 2021

- Topic: "Low-Light NeRF"

Professional Experience

Research Group Lead

GERMAN RESEARCH CENTER FOR ARTIFICIAL INTELLIGENCE (DFKI)

Leading Research projects on Image-based Rendering, VR and 3D Reconstruction

Kaiserslautern, Germany

Nov. 2023 - Now

Software Engineer

ICOG LABS

I worked on research projects in unsupervised learning from videos and single image collections

Addis-Ababa, Ethiopia

Oct. 2013 - Oct. 2014

Assistant Lecturer

AKSUM UNIVERSITY

Network Analysis and Synthesis, Electromagnetic Field Theory

Aksum, Ethiopia

Aug. 2012 - Sep. 2013

Teaching Experience

3D Computer Vision

RPTU KAISERSLAUTERN-LANDAU

Kaiserslautern, Germany

Winter Sem. 2018-2022

Graph Theory

RPTU KAISERSLAUTERN-LANDAU

Kaiserslautern, Germany

Summer Sem. 2016

Case-based Reasoning

RPTU KAISERSLAUTERN-LANDAU

Kaiserslautern, Germany

Winter Sem. 2016

Programming Languages

Python, Lua, Matlab, C, C++

CUDA, PyCUDA

Selected Publications

Text2MPI: Learning to Generate Multi-Plane Images from Text. Tewodros **Habtegebrial**, Stefan Popov, Didier Stricker, Varun Jampani and Vittorio Ferrari. Under Review, 2023.

SOMSI: Spherical Novel View Synthesis with Soft Occlusion Multi-sphere Images. Tewodros **Habtegebrial**, Christiano Gava, Marcel Rogge, Stricker Didier and Varun Jampani. CVPR, 2022.

Generative View Synthesis: From Single-view Semantics to Novel-view Images. Tewodros **Habtegebrial**, Varun Jampani, Orazio Gallo, and Didier Stricker. NeurIPS, 2020.

Fast View Synthesis with Deep Stereo Vision. Tewodros **Habtegebrial**, Kiran Varanasi, Christian Bailer, and Didier Stricker. VISAPP, 2019.

Deep Convolutional Networks For Snapshot Hyperspectral Demosaicking. Tewodros **Habtegebrial**, Gerd Reis, and Didier Stricker. WHISPERS, 2019.

Fast Feature Extraction with CNNs with Pooling Layers. Christian Bailer, Tewodros **Habtegebrial**, Kiran Varanasi, and Didier Stricker. BMVC, 2018.

SphereGlue: Learning Keypoint Matching on High Resolution Spherical Images. Christiano Gava, Vishal Mukunda, Tewodros **Habtegebrial**, Federico Raue, Sebastian Palacio and Andreas Dengel. CVPR-Workshops, 2023

Amharic Text Image Recognition: Database, Algorithm, and Analysis. Birhanu Hailu Belay, Tewodros **Habtegebrial**, Marcus Liwicki, Gebyehu Belay and Didier Stricker. International Conference on Document Analysis and Recognition (ICDAR), 2019

Factored Convolutional Neural Network For Amharic Character Image Recognition. Birhanu Hailu Belay, Tewodros **Habtegebrial**, Marcus Liwicki, Gebyehu Belay and Didier Stricker. International Conference on Image Processing (ICIP), 2019